

Machine	Temperature	Vibration	Failure
1	High	High	Yes
2	High	Low	Yes
3	Medium	High	Yes
4	Low	Low	No
5	Low	Medium	No
6	Medium	Low	No
7	High	Medium	Yes
8	Low	High	No

Failure (Variable) = $\frac{4}{8} (0) + \frac{4}{8} (1) + \frac{0}{8} (0)$

Yes = 4

No = 4

Total = 8

$$\begin{aligned}
 \text{Entropy (S)} &= - \sum P_i \log_2 P_i \\
 &= - \frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8} \\
 &= - 0.346 + 0.346 \\
 &= \underline{\underline{0.692}} = 0.5 + 0.5 \\
 &= 1
 \end{aligned}$$

Consider Attribute "Temperature",

• Temperature = Low

Yes = 0

No = 3

Entropy (Low) = 0

• Temperature = Medium

Yes = 1

No = 1

$$\text{Entropy (Medium)} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ = 1$$

• Temperature = High

Yes = 3

No = 0

$$\text{Entropy (High)} = 0$$

Weighted Entropy for Temperature

$$\text{Entropy (S, Temperature)} = \frac{3}{8} (0) + \frac{2}{8} \times 1 + \frac{3}{8} \times 0 \\ = \underline{\underline{0.25}}$$

Information gain for Temperature

$$\text{Gain (S, Temperature)} = \text{Entropy (S)} - \text{Entropy (S, Temp)} \\ = 1 - 0.25 \\ = \underline{\underline{0.75}}$$

Consider attribute - Vibration

• Vibration - Low,

Yes - 1

No - 2

$$\begin{aligned} \text{Entropy (Low)} &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \\ &= \cancel{1.584} \quad \underline{\underline{0.917}} \end{aligned}$$

• Vibration - Medium,

Yes - 1

No - 1

$$\begin{aligned} \text{Entropy (Medium)} &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= \underline{\underline{1}} \end{aligned}$$

• Vibration - High,

Yes - 2

No - 1

$$\text{Entropy (High)} = \cancel{\frac{1}{3}} \quad \cancel{1.584} \quad \underline{\underline{0.917}}$$

Weighted Entropy For ~~Ben~~ Vibration

$$\begin{aligned} \text{Entropy (S, Vibration)} &= \frac{3}{8} (\cancel{1.584})^{0.917} + \frac{2}{8} (1) + \frac{3}{8} (\cancel{1.584})^{0.917} \\ &= \cancel{1.438} \quad \underline{\underline{0.937}} \end{aligned}$$

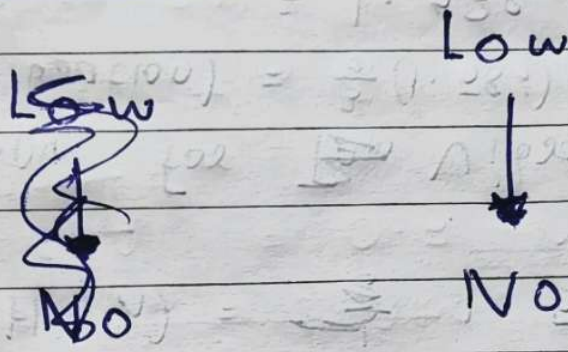
Information gain For Vibration

$$\begin{aligned} \text{Gain (S, Vibration)} &= \text{Entropy (S)} - \text{Entropy (S, vib)} \\ &= \cancel{1.438} = 1 - 0.937 \end{aligned}$$

$$= \cancel{0.438} = \underline{\underline{0.063}}$$

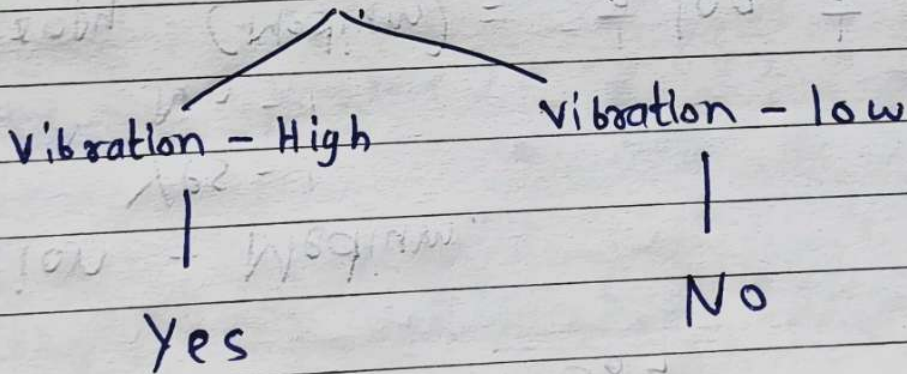
Here Temperature is the root node

Temperature = Low



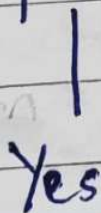
Temperature = Medium

Medium

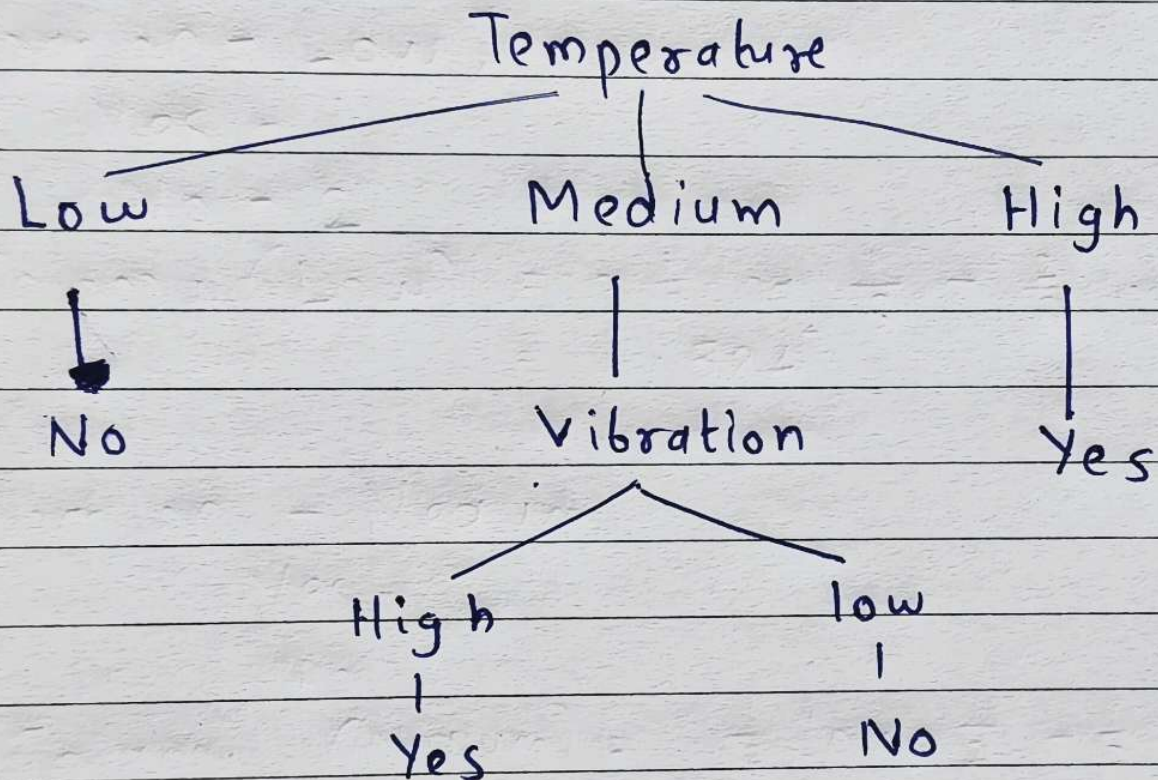


Temperature = High

Temperature



Final Decision Tree



★ Data for prediction :

Temperature = High

Vibration = Medium

According to Decision Tree

Temperature → High → Failure
Yes

The machine will Fail